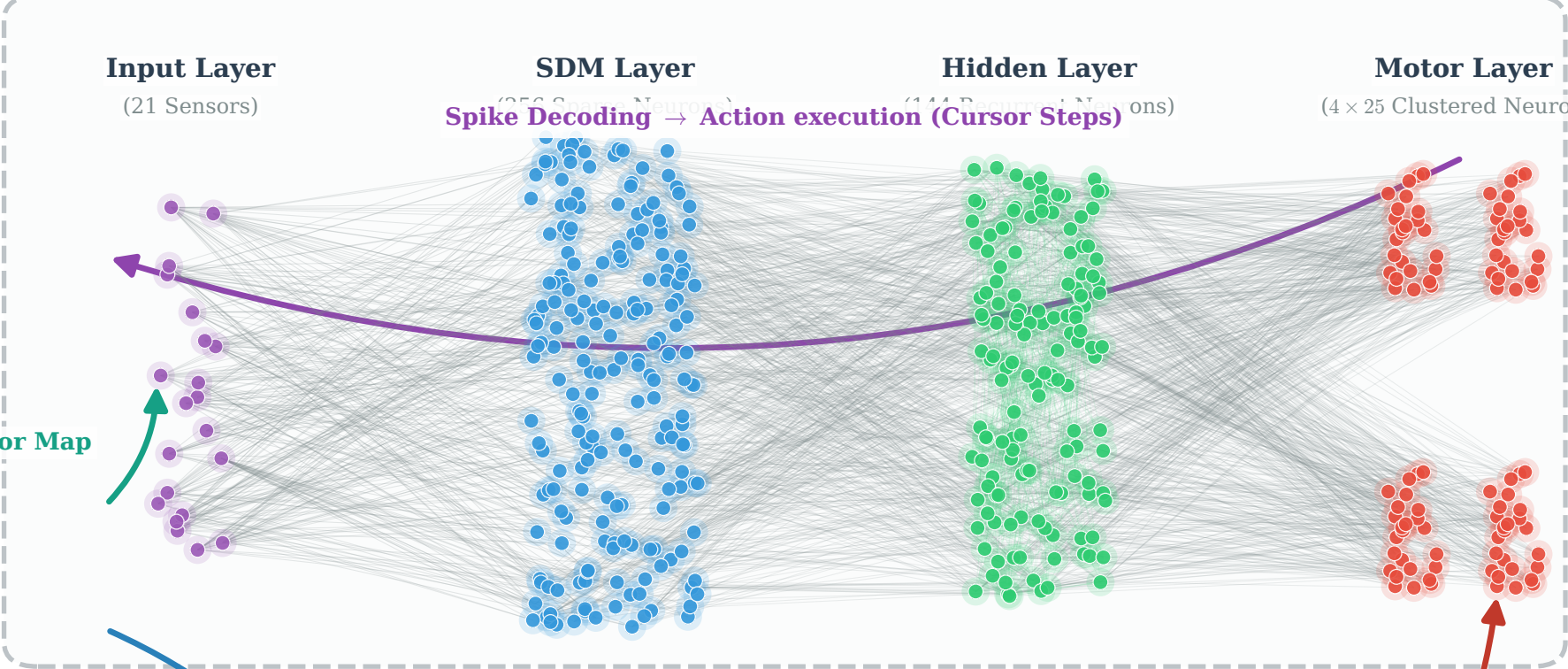


The Organoid (*in silico* Metabolic-Izhikevich Spiking Neural Network)



Exteroceptive Sensor Map

State s_t
Reward r_t

Global Activity Regulator (Homeostatic Reset) \& Inhibitory Homeostasis \& Structural Plasticity

Motor Quadrant
Injection I_{stim}

Automated Experimentalist (Outer Agent)

Dueling Double DQN + Prioritized Exp. Replay

$$Q^*(s_t, a_t) = V(s_t) + \left(A(s_t, a_t) - \frac{1}{|\mathcal{A}|} \sum A(s_t, a) \right)$$